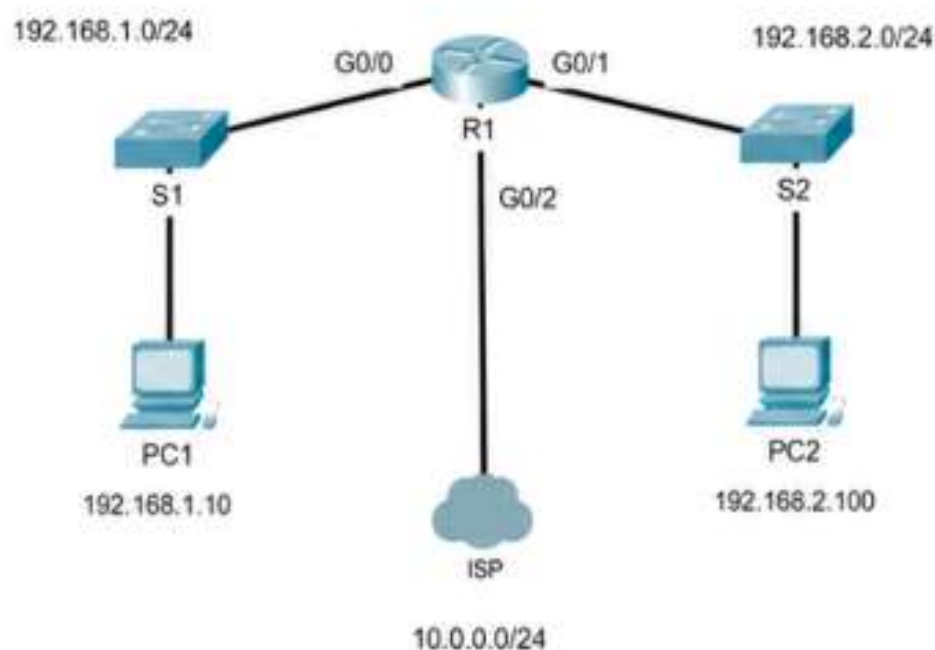


Logical Physical x: 720, y: 134

**R1 - Routing Table**

R1# show ip route

Codes: C - connected, S - static, * - candidate default, U - per-user static,
o - ODR

Gateway of last resort is 10.0.0.1 to network 0.0.0.0

```
S    192.168.1.0/24 [1/0] via 10.0.0.2
S    192.168.2.0/24 [1/0] via 10.0.0.3
S*   0.0.0.0/0 [1/0] via 10.0.0.1
```

R1#

Time: 00:16:48



R1# show ip route 192.168.2.100

Routing entry for 192.168.2.0/24

Known via "static", distance 1, metric 0

Routing Descriptor Blocks:

* 10.0.0.3, from 192.168.2.0/24, via GigabitEthernet0/1

Route metric is 0, traffic share count is 1

R1#

R1# ping 192.168.2.100

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.2.100, timeout is 2 seconds:

!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms

R1#

R1# show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.1.1	YES	manual	up	up
GigabitEthernet0/1	192.168.2.1	YES	manual	up	up
GigabitEthernet0/2	10.0.0.1	YES	manual	up	up

R1#

R1# show ip route

Codes: C - connected, S - static, * - candidate default, U - per-user static, o - ODR

Gateway of last resort is 10.0.0.1 to network 0.0.0.0

S 192.168.1.0/24 [1/0] via 10.0.0.2

S 192.168.2.0/24 [1/0] via 10.0.0.3

S* 0.0.0.0/0 [1/0] via 10.0.0.1

R1#

Ctrl+F6 to exit CLI focus

Copy

Paste

☐ Top

Realtime

Simulation

Fire Last Status Source Destination Time Color Time/call Periodic Num

1. **Matching Entry (Longest-Prefix Match)** → 192.168.2.0/24 is the best match for 192.168.2.100.
2. **Next-Hop Router Selected** → Packet will be forwarded to 10.0.0.3 via G0C/1.
3. **Destination IP Verification** → 192.168.2.100 belongs to 192.168.2.0/24 network.
4. **Possible Reasons (if packets not reaching):**
 - Check connectivity between R1 and next-hop 10.0.0.3.
 - Verify interfaces are up/up.
 - Verify host PC2 (192.168.2.100) is powered on and reachable.
 - Check for ACLs or firewall rules blocking traffic.

5. Optimization

- Static routes are simple and efficient here.
- For large networks, use dynamic routing (OSPF/EIGRP).
- Summarize routes where possible.

6. If No Matching Destination Exists

- Default route 0.0.0.0/0 via 10.0.0.1 will be used.

